THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Final Subject Assessment, 2024

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 28 pages (including this page) with 10 questions and 87 total marks

Permitted Materials

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- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. If you do not have enough space, tick the box near the bottom of page 28 and request a booklet from an invigilator—include the question number at the top of each page in the booklet.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.

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Question 1 (7 marks)

(a) Let p and q be primitive statements. Write a truth table for the compound statement

$$(q \wedge p) \wedge ((\sim q) \vee p)$$

Is this statement a contradiction? Explain.

(b) Given a compound statement, let's define a *flip* to be the operation of changing a single $\vee$ to a $\wedge$ or a single $\wedge$ to a $\vee$. For example, the statements $p \vee q$ and $p \wedge q$ can be obtained from each other by a single flip. What is the smallest number of flips needed to transform the compound statement from part (a) into a tautology? Write down your proposed statement and a truth table demonstrating it is a tautology. (You do not need to write a proof that fewer flips will not be sufficient.)

(a)

p	q	$\sim q$	$q \wedge p$	$(\sim q) \vee p$	$(q \wedge p) \wedge ((\sim q) \vee p)$
T	T	F	T	T	T
T	F	T	F	T	F
F	T	F	F	F	F
F	F	T	F	T	F

The statement $(q \wedge p) \wedge ((\sim q) \vee p)$ is true when p and q are both true; therefore, it is not a contradiction, since it is not false regardless of the truth value of the primitive statements.

(b) We need two flips. For example, the statement $(q \vee p) \vee ((\sim q) \vee p)$ is a tautology, as demonstrated by the truth table below.

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(b)

p	q	$\sim q$	$q \vee p$	$(\sim q) \vee p$	$(q \ \vee \ p) \ \vee \ ((\sim q) \ \vee \ p)$
T	T	F	T	T	T
T	F	T	T	T	T
F	T	F	T	F	T
F	F	T	F	T	T

correct truth table

Question 2 (7 marks)

Let c be a real number, $c \neq 1$. Using a proof by induction, prove that

$$\sum_{i=0}^{n} c^i = \frac{c^{n+1} - 1}{c - 1}$$

for all $n \in \mathbb{N}$.

Proof. We proceed by induction on $n \geq 0$. Let $p(n)$, $n \in \mathbb{N}$ be the condition

$$\sum_{i=0}^{n} c^i = \frac{c^{n+1} - 1}{c - 1}$$

1. *Show that $p(0)$ is true.*

   When $n = 0$,

   $$\sum_{i=0}^{n} c^i = 1 = \frac{c - 1}{c - 1},$$

   so $p(0)$ is true.

2. *Show that for each $k \in \mathbb{N}$, if $p(k)$ is true then $p(k+1)$ is true.*

   Suppose that for some $k \in \mathbb{N}$, $p(k)$ is true.

   In other words,

   $$\sum_{i=0}^{k} c^i = \frac{c^{k+1} - 1}{c - 1}.$$

   Then

   $$\begin{aligned}
   \sum_{i=0}^{k+1} c^i &= \left(\sum_{i=0}^{k} c^i\right) + c^{k+1} \\
   &= \frac{c^{k+1} - 1}{c - 1} + c^{k+1} \\
   &= \frac{c^{k+1} - 1}{c - 1} + \frac{c^{k+1}(c - 1)}{c - 1} \\
   &= \frac{c^{k+1} - 1 + c^{k+2} - c^{k+1}}{c - 1} \\
   &= \frac{c^{k+2} - 1}{c - 1}
   \end{aligned}$$

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Here the second $=$ holds by the inductive hypothesis that $p(k)$ is true, and the others are algebra.
Thus,

$$\sum_{i=0}^{k+1} c^i = \frac{c^{k+2} - 1}{c - 1} ,$$

that is, $p(k+1)$ holds true.
Therefore, by the Principle of Mathematical Induction, $p(n)$ holds true for all $n \in \mathbb{N}$.

Question 3 (4 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists x^{-1} \in \mathbb{R} \quad x \cdot x^{-1} = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Let $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$, and $c \in \mathbb{R}$, and assume $ab = ac$. Using only the Axioms of the Real Numbers (and rules of logic), write a formal proof showing that $b = c$.

Proof.

1. $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$, $c \in \mathbb{R}$	(premise)
2. $ab = ac.$	(premise)
3. $ba = ca$	(2, M2)
4. $\left(\exists a^{-1} \in \mathbb{R}\right)\ a \cdot a^{-1} = 1$	(M5)
5. $(ba) \cdot a^{-1} = (ca) \cdot a^{-1}$	(3,4)
6. $b(a \cdot a^{-1}) = c(a \cdot a^{-1})$	(5, M3)
7. $b \cdot 1 = c \cdot 1$	(6, 4)
8. $1 \cdot b = 1 \cdot c$	(7, M2)
9. $b = c$	(8, M4)

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Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 4 (7 marks)

Let $E \subset \mathbb{R}$ be a non-empty bounded below set.

(a) Write the statement "E is bounded below" in formal mathematical language.

(b) Let $F = \{x \in \mathbb{R} \mid (\forall y \in E)\, x < y\}$. Show that F is bounded above and $\sup F = \inf E$.

(a)

$$(\exists c \in \mathbb{R})\,(\forall x \in E)\, c \leq x$$

(b) Since E is non-empty, F is bounded above by any element of E. Since E is bounded below, F (which is the set of lower bounds of E) is non-empty. By completeness, F has a supremum, let's call it L.
We first show that L is a lower bound of E. Assume for a contradiction that it is not; then there exists $y \in E$ such that $y < L$. By definition of F, $x \leq y$ for all $x \in F$; thus, y is an upper bound of F, and $y < L$. This contradicts the assumption that L is the least upper bound of F.
Next, we prove that L is the *greatest* lower bound (infimum) of E. Let x be any lower bound of E, that is, $(\forall y \in E)\, x \leq y$, and let $\epsilon > 0$. Then $(\forall y \in E)\, x - \epsilon < y$; by definition, $x - \epsilon \in F$. Since $L = \sup F$, $x - \epsilon \leq L$. Because $\epsilon > 0$ is arbitrary, we conclude that $x \leq L$. We have proved that for every lower bound x of E, $x \leq L$; that is, $L = \inf E$, as asserted.

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Question 5 (8 marks)

In each part of this question, you are asked to give an example of something, and briefly explain why it is an example.

(a) A non-empty bounded subset $A \subseteq \mathbb{R}$ such that $\sup A$ is not a limit point of A.

(b) A series $\sum_{n=0}^{\infty} a_n$ such that $a_n \neq 0$ for all n, but $\sum_{n=0}^{\infty} a_n = 0$.

(c) An unbounded function $f : \mathbb{R} \to \mathbb{R}$ that is not continuous at any real number a.

(a) Any non-empty finite set A will work. Indeed, a finite set if always bounded, and its set of limit points is empty. (There are, of course, many infinite examples as well - for example, strictly monotone decreasing bounded sequences, or unions of any bounded set E with a finite set that contains an upper bound for E larger than $\sup E$, although the arguments tend to be more complicated for these examples.)

(b) Use the algebra of limits (or series) theorem. For example, let $b_n = 1/2^n$, so the series $\sum_{n=0}^{\infty} b_n$ converges to 2; now subtract 2 from one of the terms. So, for example, consider the series such that $a_0 = -1$ and $a_n = 1/2^n$ for $n > 0$.

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(c) Consider the function $f(x)$ defined as $f(x) = 0$ if x is rational, and $f(x) = 1/x$ if x is irrational. Then f is unbounded (for example, $f(\sqrt{2}/n) = n/\sqrt{2}$, which is unbounded by the Archimedean Principle). The function $g(x) = 1 - xf(x)$ takes value 1 at all rational numbers, and value 0 at all irrational numbers; in other words, it is the rational indicator function, which was shown to be nowhere continuous in tutorial. But the functions 1 and x are continuous, so the algebra of continuous functions theorem implies $f(x)$ cannot be continuous anywhere.

(Note that there are, of course, lots of examples. Eg one could simply modify the rational indicator functions by setting $f(x) = \lceil x \rceil$ if x is rational and say that one can prove it is nowhere continuous similarly to the argument for the rational indicator function.)

Question 6 (11 marks)

Consider the sequence (f_n) defined inductively by $f_1 = 1$ and, for all $n \geq 1$,

$$f_{n+1} = \frac{f_n}{2} + \frac{1}{f_n}.$$

(a) Using induction on $n \geq 1$, show that, for all $n \in \mathbb{N}^+$, $f_n \geq 1$.

(b) Using part (a), show that, for all $n \in \mathbb{N}^+$,

$$|f_{n+2} - f_{n+1}| \leq \frac{1}{2} |f_{n+1} - f_n| .$$

(**Hint:** Do not attempt a proof by induction for this part.)

(c) Using part (b) and results from lectures, prove that $\lim_{n\to\infty} f_n = \sqrt{2}$.

(a) We proceed by induction on $n \geq 1$. For $n = 1$, we have $f_n = 1 \geq 1$ by definition. Assume now that, for some $k \geq 1$, $f_k \geq 1$. Then

$$f_{k+1} - 1 = \frac{f_k}{2} + \frac{1}{f_k} - 1 = \frac{f_k^2 - 2f_k + 2}{2f_k} = \frac{(f_k - 1)^2 + 1}{2f_k} \geq 0$$

where the final inequality holds because $f_k > 0$ by inductive hypothesis, and since squares are always non-negative. Thus, $f_{k+1} \geq 1$. By the principle of mathematical induction, we conclude that $f_n \geq 1$ for all $n \in \mathbb{N}^+$. (Alternatively, one can distinguish between the cases $f_k \geq 2$ and $1 \leq f_k < 2$. In the latter case, both $f_k/2$ and $1/f_k$ are $\geq 1/2$.)

(b) We compute

$$f_{n+2} - f_{n+1} = \frac{1}{2}(f_{n+1} - f_n) + \left(\frac{1}{f_{n+1}} - \frac{1}{f_n}\right) =$$

$$\frac{1}{2}(f_{n+1} - f_n) + \frac{1}{f_n f_{n+1}}(f_n - f_{n+1}) = \left(\frac{1}{2} - \frac{1}{f_n f_{n+1}}\right) \cdot (f_{n+1} - f_n)$$

By part (a), $f_n f_{n+1} \geq 1$, so that

$$-\frac{1}{2} \leq \frac{1}{2} - \frac{1}{f_n f_{n+1}} \leq \frac{1}{2}$$

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and thus

$$\left|\frac{1}{2} - \frac{1}{f_n f_{n+1}}\right| \leq \frac{1}{2}$$

Therefore,

$$|f_{n+2} - f_{n+1}| = \left|\frac{1}{2} - \frac{1}{f_n f_{n+1}}\right| \cdot |f_{n+1} - f_n| \leq \frac{1}{2}|f_{n+1} - f_n|.$$

(c) By part (b), the sequence is contractive and therefore converges to some limit L. Multiplying the defining inductive relation by the non-zero number $2f_n$, we see that

$$2f_n f_{n+1} = f_n^2 + 2.$$

Since (f_n) converges so does (f_{n+1}), and to the same limit; therefore we may apply algebra of limits and conclude that $2L^2 = L^2 + 2$, or $L^2 = 2$. (Note: if you want to apply algebra of limits before eliminating the denominator, you have to first prove that $L \neq 0$ separately.) Finally, it follows from part (a) that $L \geq 1$ and in particular, positive. Thus, $L = \sqrt{2}$.

Question 7 (10 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{1}{x^2+2}.$$

(a) Using the ϵ - δ definition of limits, show that $\lim_{x\to 0} f(x) = 1/2$. Write your scrap work as well as proof.

(b) Using the ϵ - M definition of limits, show that $\lim_{x\to\infty} f(x) = 0$. Write your scrap work as well as proof.

(c) Let $0 < c < 1/2$. Without solving for x, prove that there is a real number x such that $f(x) = c$. (For this part only, you may use without proof that $f(x)$ is continuous.)

(a) **Scrap work.** Let $\epsilon > 0$. Then

$$\left|\frac{1}{x^2+2} - \frac{1}{2}\right| < \epsilon$$
$$\left|\frac{-x^2}{2x^2+4}\right| < \epsilon$$

Since $2x^2 + 4 \geq 4$, this estimate will hold provided $x^2 < 4\epsilon$, or $|x| < 2\sqrt{\epsilon}$. Thus, we can take $\delta = 2\sqrt{\epsilon}$.

Proof. Let $\epsilon > 0$ and let $\delta = 2\sqrt{\epsilon}$. If $0 < |x| < \delta$, we have

$$\left|\frac{1}{x^2+2} - \frac{1}{2}\right| = \left|\frac{-x^2}{2x^2+4}\right| \leq \frac{1}{4}\left|x^2\right| < \frac{1}{4}\delta^2 = \epsilon.$$

Thus, if $0 < |x| < \delta$, then $\left|f(x) - \frac{1}{2}\right| < \epsilon$. That is, $\lim_{x\to 0} f(x) = 1/2$.

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(b) **Scrap work.** Let $\epsilon > 0$. Then

$$\begin{aligned} \left|\frac{1}{x^2+2} - 0\right| &< \epsilon \\ \left|\frac{1}{x^2+2}\right| &< \epsilon \\ \frac{1}{\epsilon} &< 2 + x^2 \\ \frac{1}{\epsilon} - 2 &< x^2 \end{aligned}$$

Let $M = \sqrt{\frac{1}{\epsilon}}$.

Proof. Let $\epsilon > 0$ and let $M = \sqrt{\frac{1}{\epsilon}} > 0$. If $x > \sqrt{\frac{1}{\epsilon}}$, we have

$$\left|\frac{1}{x^2+2}\right| = \frac{1}{x^2+2} < \frac{1}{2+\frac{1}{\epsilon}} < \frac{1}{\frac{1}{\epsilon}} = \epsilon.$$

Thus, if $x > M$, then $|f(x) - 0| < \epsilon$. That is, $\lim_{x\to\infty} f(x) = 0$.

(c) Let $0 < c < 1/2$. By part (b), there exists $a > 0$ such that $0 < f(a) < c$. (Take $\epsilon = c$, and $a > M$.) Thus, $f(0) > c > f(a)$. Since $f(x)$ is continuous on $[0, a]$, the Intermediate Value Theorem implies that there exists $x \in [0, a]$ such that $f(x) = c$.

Question 8 (12 marks)

Let $a < b$ and $f : [a, b] \to \mathbb{R}$ be a bounded function.

(a) Let $P = \{a = x_0 < x_1 < x_2 < \cdots < x_n = b\}$ be a partition. Give the general expression of the lower Riemann sum $L(f, P)$ and upper Riemann sum $U(f, P)$, defining all your terms.

(b) Continuing from part (a), assume in addition that f is monotone increasing. For $i = 1, \ldots, n$, show that

$$\sup\{f(x) \mid x \in [x_{i-1}, x_i]\} = f(x_i) \text{ and } \inf\{f(x) \mid x \in [x_{i-1}, x_i]\} = f(x_{i-1}).$$

(c) Let $\delta > 0$. Continuing from part (b), assume moreover that $x_i - x_{i-1} < \delta$ for all $i = 1, \ldots, n$. Show that

$$U(f, P) - L(f, P) < \delta \cdot (f(b) - f(a)).$$

(d) Using part (c), prove: If $f : [a, b] \to \mathbb{R}$ is a bounded monotone increasing function, then for all $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$. (It follows - but you do not need to prove - that f is Riemann integrable.)

(a) Let $m_i = \inf\{f(x) \mid x \in [x_{i-1}, x_i]\}$ and $M_i = \sup\{f(x) \mid x \in [x_{i-1}, x_i]\}$. Then

$$L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1})$$

and

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1})$$

correct expressions, and all terms defined.

(b) Since f is monotone increasing, $f(x_{i-1}) \leq f(x) \leq f(x_i)$ for all $x \in [x_{i-1}, x_i]$. Thus, $f(x_{i-1})$ is a lower, and $f(x_i)$ an upper bound for $\{f(x) \mid x \in [x_{i-1}, x_i]\}$. On the other hand, x_{i-1} and x_i are points in the interval $[x_{i-1}, x_i]$; it follows that $f(x_{i-1})$ is a greatest lower bound (in fact, minimum) and $f(x_i)$ a least upper bound (in fact, a maximum).

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(c) From parts (a) and (b), we compute

$$U(f,P) - L(f,P) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) =$$

$$= \sum_{i=1}^{n} \left(f(x_i) - f(x_{i-1})\right)(x_i - x_{i-1}).$$

Since f is monotone increasing, all the terms $f(x_i) - f(x_{i-1})$ are non-negative; therefore the hypothesis that $x_i - x_{i-1} < \delta$ implies that

$$\sum_{i=1}^{n} \left(f(x_i) - f(x_{i-1})\right)(x_i - x_{i-1}) < \delta \sum_{i=1}^{n} \left(f(x_i) - f(x_{i-1})\right).$$

The last sum is a telescope sum:

$$\sum_{i=1}^{n} \left(f(x_i) - f(x_{i-1})\right) = f(b) - f(a).$$

Putting everything together, we have proved what we wanted:

$$U(f,P) - L(f,P) < \delta \cdot \left(f(b) - f(a)\right).$$

(d) If $f(b) = f(a)$, then f is constant of some value v and $L(f,P) = U(f,P) = v(b-a)$ for all partitions P. Now assume $f(b) \neq f(a)$. Let $\delta = \epsilon/(f(b) - f(a))$. Using the Archimedean Principle, we can find $n \in \mathbb{N}^+$ such that $(b-a)/n < \delta$. Let P be the partition with

$$x_i = a + \frac{i(b-a)}{n}.$$

Then P satisfies the hypothesis of part (c), so we conclude that

$$U(f,P) - L(f,P) < \delta \cdot \left(f(b) - f(a)\right) = \epsilon.$$

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Note. *For Questions 9 and 10, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 9 (13 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{\log(n+1)}{(n+1)^2}$

(ii) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{1+n} - \frac{1}{n}\right)$

(b) Find the radius of convergence and the interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{n^4 \cdot x^n}{2^n}.$$

(a) (i) Converges.

Let $g(x) = \frac{\log x}{x^2}$. The derivative is $g'(x) = \frac{(1-2\log x)x}{x^4}$, which is negative for $x \geq 2$ (because $x^2 > e$.) We conclude that the function $f(x) = g(x+1)$ is montone decreasing for $x \geq 1$; we also have $f(x) > 0$ for all $x \geq 1$. Thus, the integral test applies. The anti-derivative $G(x)$ of $g(x)$ can be computed using integration by parts: $G(x) = -\frac{1+\log x}{x}$.
Therefore,

$$\lim_{x\to\infty} \int_1^x f(t)dt = \lim_{x\to\infty} \left(-\frac{1+\log(x+1)}{x+1} + \frac{1+\log 2}{2}\right) = \frac{1+\log 2}{2}$$

(Alternatively, one can use comparison with a p-harmonic series for some $1 < p < 2$.)

(ii) Converges.

We compute

$$a_n = \frac{1}{n+1} - \frac{1}{n} = -\frac{1}{n^2+n}.$$

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The series converges absolutely by comparison with the 2-harmonic series as $\frac{1}{n^2+n} < \frac{1}{n^2}$.

(b) By the Ratio Test, since

$$\begin{aligned} r &= \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} \\ &= \lim_{n\to\infty} \frac{(n+1)^4|x|^{n+1}}{2^{n+1}} \cdot \frac{2^n}{n^4|x|^n} \\ &= \lim_{n\to\infty} \frac{(n+1)^4\,|x|}{2n^4} \\ &= \frac{|x|}{2}, \end{aligned}$$

the power series converges if $r = |x|/2 < 1$, i.e. if $|x| < 2$. Moreover, the power series diverges if $r = |x|/2 > 1$, i.e. if $|x| > 2$. In particular, the radius of convergence is 2.

At the endpoint $x = -2$, we obtain the series

$$\sum_{n+1}^{\infty} (-1)^n n^4$$

which diverges by the divergence test.

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At the endpoint $x = 2$, we obtain the series

$$\sum_{n+1}^{\infty} n^4$$

which also diverges by the divergence test.

Therefore, the interval of convergence is $(-2, 2)$.

Question 10 (8 marks)

Let $f(x) = \sin(x) - x$.

(a) Compute the degree 4 Taylor polynomial of $T_4(x)$ of $f(x)$.

(b) Use Taylor's Theorem to show that, for $x \in \mathbb{R}$

$$|f(x) - T_4(x)| \leq \left|\frac{x^5}{120}\right|.$$

(c) Suppose $|x| \leq 0.1$. To how many digits (at least) does $T_4(x) + x$ approximate $\sin x$?

(a) The first four derivatives of $f(x)$ are $\cos x - 1$, $-\sin x$, $-\cos x$, $\sin x$. Evaluating at 0, we obtain

$$T_4(x) = -\frac{x^3}{6}$$

(b) We compute the fifth derivative $f^{(5)}(x) = \cos x$. By Taylor's Theorem, there exists c between 0 and x such that

$$f(x) - T_4(x) = \frac{f^{(5)}(c)}{120} \cdot x^5$$

so by our calculation

$$|f(x) - T_4(x)| = \left|\frac{\cos(c)}{120} \cdot x^5\right| = \left|\frac{\cos(c)}{120}\right| \cdot |x^5|$$

Since $|\cos c| \leq 1$, we obtain the desired estimate.

(c) Using part (b), we estimate

$$|T_4(x) + x - \sin x| \leq \frac{(0.1)^5}{120} \leq 10^{-7}.$$

Thus, the approximation is to (at least) six digits.

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End of Exam — Total Available Marks = 87

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